

Individual Contributions

The Smart Greenhouse project was carried out as a team effort, with each member taking responsibility for a particular part of the project while also helping in other areas whenever needed. We worked together throughout the design, assembly, programming, testing, and documentation stages, discussing problems and making decisions as a group.

Walid Al Basha contributed mainly to the Arduino programming, system integration, control logic, and technical documentation.

Mohammad Fakh contributed mainly to the electrical wiring, hardware assembly, sensor connections, and testing of the electronic components.

Nancy Mezher contributed mainly to sensor calibration, data organization, report writing, and analysis of the experimental results.

Ahmad Gabris contributed mainly to the mechanical assembly of the greenhouse prototype, water distribution system, system testing, and preparation of the project demonstration.

Although each member had a primary responsibility, everyone participated in troubleshooting, improving the system, preparing the GitHub repository, reviewing the report, and recording the demonstration and technical presentation. The final prototype was the result of the group's combined effort and cooperation.

This project gave us the opportunity to apply concepts learned in class to a real engineering system. Along the way, we discovered that building a working prototype involves much more than simply connecting components and writing code. Small issues such as wiring mistakes, sensor calibration, or unexpected software behavior often required patience and teamwork to solve. Working together helped us improve our technical skills while also developing better communication, planning, and problem-solving abilities. Seeing the final system operate successfully made the effort worthwhile and gave us a better understanding of how different engineering disciplines come together in a practical project.